

NAMA : FAFIQ LUTFI AZANA
KELAS : TI-1G
NIM ; 254107020058

LAPORAN TUGAS JOBSHEET 3

```
1. package PraktikumAlgoritmaDanStrukturData.jobsheet3.Latihan;  
import java.util.Scanner;
```

```
public class Dosen09 {  
    String kode;  
    String nama;  
    boolean jenisKelamin;  
    int usia;
```

```
    public Dosen09(String kode, String nama, boolean jenisKelamin, int usia) {  
        this.kode = kode;  
        this.nama = nama;  
        this.jenisKelamin = jenisKelamin;  
        this.usia = usia;  
    }
```

```
    void tampil() {  
        System.out.println("Kode : " + kode);  
        System.out.println("Nama : " + nama);  
        System.out.println("Jenis Kelamin : " + (jenisKelamin ? "Pria" : "Wanita"));  
        System.out.println("Usia : " + usia);  
        System.out.println("=====");  
    }
```

```
    void inputData(Scanner sc) {  
        System.out.print("Kode dosen\t: ");  
        kode = sc.nextLipackage PraktikumAlgoritmaDanStrukturData.jobsheet3.Latihan;  
import java.util.Scanner;
```

```
public class Dosen09 {  
    String kode;  
    String nama;  
    boolean jenisKelamin;  
    int usia;
```

```
    public Dosen09(String kode, String nama, boolean jenisKelamin, int usia) {
```

```

        this.kode = kode;
        this.nama = nama;
        this.jenisKelamin = jenisKelamin;
        this.usia = usia;
    }

```

```

    void tampil() {
        System.out.println("Kode      : " + kode);
        System.out.println("Nama      : " + nama);
        System.out.println("Jenis Kelamin : " + (jenisKelamin ? "Pria" : "Wanita"));
        System.out.println("Usia      : " + usia);
        System.out.println("=====");
    }

```

```

    void inputData(Scanner sc) {
        System.out.print("Kode dosen\t: ");
        kode = sc.nextLine();
        System.out.print("Nama dosen\t: ");
        nama = sc.nextLine();
        System.out.print("Jenis Kelamin (true = pria, false = wanita): ");
        jenisKelamin = sc.nextBoolean();
        System.out.print("Usia\t: ");
        usia = sc.nextInt();
        sc.nextLine();
    }

```

```

}

```

```

System.out.print("Nama dosen\t: ");

```

```

nama = sc.nextLine();

```

```

System.out.print("Jenis Kelamin (true = pria, false = wanita): ");

```

```

jenisKelamin = sc.nextBoolean();

```

```

System.out.print("Usia\t: ");

```

```

usia = sc.nextInt();

```

```

sc.nextLine();

```

```

}

```

```

}

```

```

2. package PraktikumAlgoritmaDanStrukturData.jobsheet3.Latihan;

```

```

import java.util.Scanner;

```

```

public class DosenDemo {

```

```

    public static void main(String[] args) {

```

```
Scanner sc = new Scanner(System.in);  
int jumlahDosen;
```

```
System.out.println("=====");  
System.out.print("Masukan jumlah dosen: ");  
jumlahDosen = sc.nextInt();  
sc.nextLine();
```

```
Dosen09[] listDosen = new Dosen09[jumlahDosen];
```

```
for (int i = 0; i < jumlahDosen; i++) {  
listDosen[i] = new Dosen09("", "", false, 0);  
}
```

```
int counter = 1;  
for (Dosen09 dosen : listDosen) {  
System.out.println("Input data dosen ke-" + counter);  
dosen.inputData(sc);  
counter++;  
System.out.println();  
}
```

```
DataDosen09 dataDosen = new DataDosen09();  
dataDosen.dataSemuaDosen(listDosen);  
dataDosen.jumlahDosenPerJenisKelamin(listDosen);  
dataDosen.rerataUsiaDosenPerJenisKelamin(listDosen);  
dataDosen.infoDosenPalingTua(listDosen);  
dataDosen.infoDosenPalingMuda(listDosen);  
  
}  
}
```

```
3. package PraktikumAlgoritmaDanStrukturData.jobsheet3.Latihan;
```

```
public class DataDosen09 {  
void dataSemuaDosen (Dosen09[] arrayOfDosen) {  
int counter = 1;  
System.out.println("===Data Semua Dosen===");  
for (Dosen09 dosen : arrayOfDosen) {  
System.out.println("Data dosen ke-" + counter);
```

```
dosen.tampil();  
counter++;  
}  
}
```

```
void jumlahDosenPerJenisKelamin(Dosen09[] arrayOfDosen) {  
int jumlahDosenPria = 0;  
int jumlahDosenWanita = 0;
```

```
for (Dosen09 dosen : arrayOfDosen) {  
if (dosen.jenisKelamin) {  
jumlahDosenPria++;  
} else {  
jumlahDosenWanita++;  
}  
}
```

```
System.out.println("=====");  
System.out.println("Jumlah dosen berdasarkan jenis kelamin");  
System.out.println("Jumlah dosen pria: " + jumlahDosenPria);  
System.out.println("Jumlah dosen wanita: " + jumlahDosenWanita);  
System.out.println();  
}
```

```
void rerataUsiaDosenPerJenisKelamin(Dosen09[] arrayOfDosen) {  
int jumlahDosenPria = 0;  
int jumlahDosenWanita = 0;  
int totalUsiaPria = 0;  
int totalUsiaWanita = 0;
```

```
for (Dosen09 dosen : arrayOfDosen) {  
if (dosen.jenisKelamin) {  
jumlahDosenPria++;  
totalUsiaPria += dosen.usia;  
} else {  
jumlahDosenWanita++;  
totalUsiaWanita += dosen.usia;  
}  
}
```

```
double rataDosenPria = (double) totalUsiaPria / jumlahDosenPria;  
double rataDosenWanita = (double) totalUsiaWanita / jumlahDosenWanita;
```

```
System.out.println("=====");
System.out.printf("Rata-rata usia dosen pria: %.2f tahun%n", rataDosenPria);
System.out.printf("Rata-rata usia dosen wanita: %.2f tahun %n", rataDosenWanita);
}
```

```
void infoDosenPalingTua(Dosen09[] arrayOfDosen){
if (arrayOfDosen.length == 0) {
System.out.println("=====");
System.out.println("Data dosen kosong!");
return;
}
```

```
Dosen09 dosenTertua = arrayOfDosen[0];
for (Dosen09 dosen : arrayOfDosen) {
if (dosen.usia > dosenTertua.usia) {
dosenTertua = dosen;
}
}
System.out.println("=====");
System.out.println("=== Data dosen tertua ===");
dosenTertua.tampil();
}
```

```
void infoDosenPalingMuda(Dosen09[] arrayOfDosen) {
if (arrayOfDosen.length == 0) {
System.out.println("=====");
System.out.println("Data dosen kosong!");
return;
}
```

```
Dosen09 dosenTerMuda = arrayOfDosen[0];
for (Dosen09 dosen : arrayOfDosen) {
if (dosen.usia < dosenTerMuda.usia) {
dosenTerMuda = dosen;
}
}
System.out.println("=====");
System.out.println("=== Data dosen termuda ===");
dosenTerMuda.tampil();
}
```

